

Problem-Solution Fit

1. Mapping Farmer Pains to Opti-Crop Features

The table below illustrates how the specific challenges faced by smallholder farmers are addressed by the design features of Opti-Crop.

Identified Farmer Pain Point	Proposed App Feature	Value Generated
Difficulty choosing crops due to unpredictable weather patterns.	Soil & Climate Predictor	Provides accurate weather forecasts and crop recommendations.
Soil degradation caused by blind application of urea/potash.	NPK Parameter Analysis	Analyzes soil nutrient levels and provides precise fertilizer recommendations.
Complex, academic scientific reports from soil labs.	Actionable Advice Generator	Translates complex scientific data into simple, actionable advice for farmers.
Losing records of previous soil health evaluations.	User Search Logs Dashboard	Saves historical test results and provides a searchable log of past evaluations.

2. Product-Market Fit Validation

Opti-Crop targets a critical niche: bridging the gap between sophisticated data science (ML classifiers) and basic field farming. By making the app lightweight and local-first, it remains usable in low-connectivity environments where access to cloud computing is restricted, locking in a strong fit for rural agricultural markets.